

Chapter 3

The Common Emitter Amplifier

Core Topology, Emitter Degeneration, Early Effect, & Biasing

*Based on Razavi, Fundamentals of Microelectronics, 2nd Ed.
Chapter 5.3.1 (Common Emitter Stage)*

With insights from Razavi's BJT YouTube Lectures (21–25)

Topics Covered:

- Review: From biasing to amplification
- The basic CE amplifier topology and voltage gain
- Small-signal analysis of the CE stage
- Input and output impedance
- Emitter degeneration: gain, linearity, and impedance trade-offs
- The “black box” perspective and G_m
- Effect of base resistance on gain
- Transistor output resistance (r_O) and the boosted output impedance
- CE stage with biasing: coupling capacitors and complete analysis
- Design methodology and worked examples

Chapter 3: Quick Reference Formula Sheet

Core CE Amplifier (No Degeneration)

Parameter	Formula	Notes
Voltage gain	$A_v = -g_m R_C$	Inverting; magnitude = $g_m R_C$
With Early effect	$A_v = -g_m (R_C \parallel r_O)$	r_O in parallel with R_C
Intrinsic gain	$ A_v = g_m r_O = V_A/V_T$	Independent of I_C !
Input impedance	$R_{in} = r_\pi = \beta/g_m$	Looking into base
Output impedance	$R_{out} = R_C$ (no r_O)	Looking into collector
Output impedance	$R_{out} = R_C \parallel r_O$ (with r_O)	Early effect lowers R_{out}

CE with Emitter Degeneration

Parameter	Formula	Notes
Voltage gain	$A_v = \frac{-g_m R_C}{1 + g_m R_E}$	Gain reduced by $1 + g_m R_E$
Approx. gain	$A_v \approx -R_C/R_E$	If $g_m R_E \gg 1$
With R_B	$A_v \approx \frac{-R_C}{1/g_m + R_E + R_B/(\beta + 1)}$	Base resistance degrades gain
Input impedance	$R_{in} = r_\pi + (\beta + 1)R_E$	R_E “multiplied” by $\beta + 1$
Output impedance	$R_{out} = R_C$ (no r_O)	Same as without R_E
Effective G_m	$G_m \approx \frac{g_m}{1 + g_m R_E}$	Effective transconductance

Output Resistance with Early Effect and Degeneration

Parameter	Formula	Notes
General R_{out}	$r_O + (g_m r_O + 1)(R_E \parallel r_\pi)$	Collector-side impedance
If $R_E \gg r_\pi$	$R_{out} \approx r_O(1 + g_m r_\pi) \approx \beta r_O$	Maximum boost
If $R_E \ll r_\pi$	$R_{out} \approx (1 + g_m R_E)r_O$	Boosted by $1 + g_m R_E$
Circuit R_{out}	$R_{out, \text{circuit}} = R_{out, \text{transistor}} \parallel R_C$	R_C always in parallel

1 Review: From Bias Point to Amplification

In Chapters 1 and 2, we established the BJT as a voltage-controlled current source and learned how to set its DC operating point. We can now summarize the **complete workflow** for analyzing any BJT amplifier:

1. **DC Analysis:** Find I_C , V_{CE} , I_B . Verify active mode.
2. **Compute small-signal parameters:** $g_m = I_C/V_T$, $r_\pi = \beta/g_m$, $r_O = V_A/I_C$.
3. **Small-signal analysis:** Replace the transistor with its model, short V_{CC} to ground, and solve for gain and impedances.

This chapter focuses entirely on Step 3 for the most fundamental amplifier topology: the **Common Emitter (CE) stage**. The name “common emitter” means the emitter terminal is *common* (shared) between the input and output ports — typically connected to ground.

CONCEPT: Why CE is the Most Important Topology

The CE amplifier is the “workhorse” of analog electronics because it provides:

- **Voltage gain:** Often the highest gain of the three topologies (CE, CB, CC)
- **Current gain:** β from base to collector
- **Phase inversion:** Output is inverted relative to input
- **Moderate input impedance:** r_π (typically 1–10 k Ω)

Understanding CE deeply is *essential* — the Common Base and Common Collector topologies in Chapters 4 and 5 are analyzed by comparing and contrasting with CE.

LECTURE INSIGHT (Lecture 21)

Razavi opens his CE amplifier lecture: “We are now ready to use everything we’ve learned — the exponential I/V, the small-signal model, biasing — and put it all together into an actual amplifier. The common emitter topology is where it all comes together.”

2 The Basic Common Emitter Stage

2.1 Circuit Topology and Physical Intuition

The basic CE stage consists of an NPN transistor with a collector resistor R_C connected to V_{CC} , the emitter tied to ground, and the input signal applied to the base:

from the collector, respectively) so as to avoid confusion in more complex topologies.

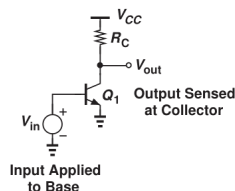


Figure 5.29 Common emitter stage

Figure 1: Figure 5.29 from Razavi — (a) Basic CE stage: R_C converts the collector current variations into an output voltage at node Y. (b) Conceptual equivalent showing the BJT as a voltage-controlled current source.

Physical intuition: Imagine a small positive signal v_{in} applied to the base.

1. The base-emitter voltage increases by v_{in} , so V_{BE} rises.
2. The collector current increases: $\Delta I_C = g_m \cdot v_{in}$ (from the transconductance).
3. This *increased* current flows through R_C , creating a *larger* voltage drop $\Delta I_C \cdot R_C$.
4. Therefore $V_{out} = V_{CC} - I_C R_C$ **decreases**. The output moves opposite to the input.
5. **Result:** The amplifier is *inverting* — $A_v < 0$.

CONCEPT: Why the CE Stage Inverts

The inversion arises because R_C sits between V_{CC} and the collector. When I_C goes up, the voltage drop across R_C increases, pulling the output node *down*. This is just Ohm's law: $V_{out} = V_{CC} - I_C R_C$. More current $\rightarrow$ more drop $\rightarrow$ lower output. The resistor “converts” current increases into voltage *decreases*.

2.2 Small-Signal Voltage Gain

current, shown in Fig. 5.30, as explained in Chapter 4, the supply voltage node, V_{CC} , ac



Figure 5.30 Small-signal model of CE stage

Figure 2: Figure 5.30 from Razavi — Small-signal equivalent of the CE stage. V_{CC} is replaced with ac ground. The controlled source $g_m v_\pi$ drives current through R_C .

DERIVATION: CE Stage Voltage Gain (No Early Effect)

In the small-signal model (Fig. 5.30):

- $v_\pi = v_{in}$ (the input voltage appears directly across r_π)
- The controlled source produces current $g_m v_\pi = g_m v_{in}$ flowing downward through R_C
- Since V_{CC} is ac ground, the output voltage at the collector is:

$$v_{out} = -g_m v_\pi \cdot R_C = -g_m v_{in} \cdot R_C$$

Therefore:

$$A_v = \frac{v_{out}}{v_{in}} = -g_m R_C$$

The **negative sign** confirms the inverting nature. The magnitude $|A_v| = g_m R_C$ shows that gain is the product of transconductance and load resistance.

Substituting $g_m = I_C/V_T$:

$$|A_v| = \frac{I_C R_C}{V_T} = \frac{I_C R_C}{26 \text{ mV}}$$

This reveals a powerful insight: the voltage gain equals the **DC voltage drop across R_C** (which is $I_C R_C$) divided by the thermal voltage V_T . For a typical drop of 1 V: $|A_v| = 1 \text{ V}/26 \text{ mV} \approx 38$.

EXAM TIP

Quick gain estimate: If you know the DC operating point, the CE gain (without degeneration) is approximately:

$$|A_v| \approx \frac{V_{R_C}}{26 \text{ mV}}$$

where V_{R_C} is the DC voltage drop across the collector resistor. This is a **very fast** exam shortcut.

2.3 Large-Signal Behavior and I/O Characteristics

The input-output characteristic reveals the **limits of linear operation**:

The I/O curve has three regions:

- **Left (cutoff):** $V_{in} < V_{BE,on} \approx 0.7 \text{ V}$. No current flows; $V_{out} = V_{CC}$.
- **Center (active):** The transistor amplifies. The slope $\partial V_{out}/\partial V_{in}$ equals $-g_m R_C$.
- **Right (saturation):** V_{out} drops to $\sim 0.2 \text{ V}$. The transistor is saturated and can no longer amplify.

The **allowable output swing** is from $V_{CE,sat} \approx 0.2 \text{ V}$ up to V_{CC} . For linear operation, we design the Q-point at the *center* of this range.

2.4 Including the Early Effect

When the transistor has a finite Early voltage V_A , the output resistance $r_O = V_A/I_C$ appears in parallel with R_C :

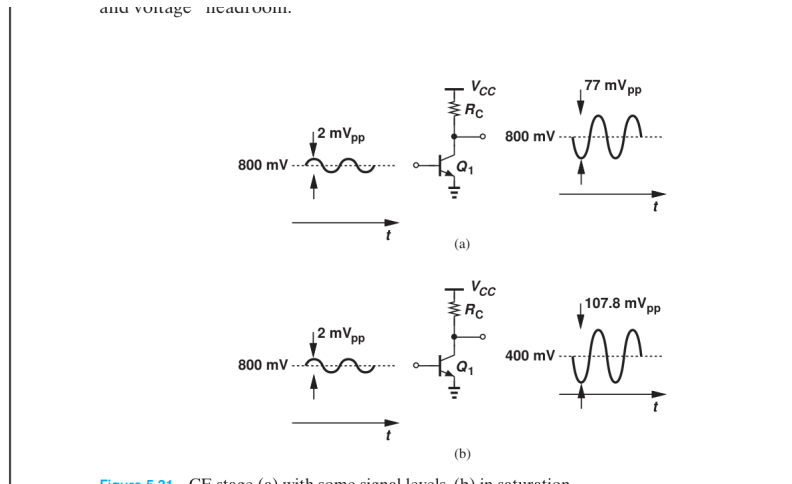


Figure 5.31 CE stage (a) with some signal levels (b) in saturation

Figure 3: Figure 5.31 from Razavi — I/O characteristic of the CE stage. The steepest slope (maximum gain) occurs in the active region. The transistor clips at both extremes: cutoff ($V_{out} \rightarrow V_{CC}$) and saturation ($V_{out} \rightarrow V_{CE,sat}$).

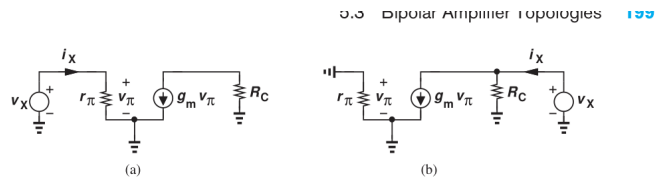


Figure 5.32 (a) Input and (b) output impedance calculation of CE stage

Figure 4: Figure 5.32 from Razavi — CE stage including r_O in the small-signal model. The output resistance of the transistor appears in parallel with R_C .

DERIVATION: CE Gain with Early Effect

With r_O included, the collector current $g_m v_\pi$ splits between R_C and r_O in parallel:

$$v_{out} = -g_m v_\pi (R_C \parallel r_O)$$

$$A_v = -g_m (R_C \parallel r_O)$$

Since typically $r_O \gg R_C$ (e.g., $r_O = 100 \text{ k}\Omega$ vs. $R_C = 1 \text{ k}\Omega$), the Early effect makes only a small correction:

$$A_v \approx -g_m R_C \left(\frac{r_O}{R_C + r_O} \right) \approx -g_m R_C$$

The Intrinsic Gain: A Fundamental Limit

What happens if we remove R_C entirely (or let $R_C \rightarrow \infty$)? The gain becomes:

$$A_v = -g_m r_O$$

Substituting $g_m = I_C/V_T$ and $r_O = V_A/I_C$:

$$|A_{v,\text{intrinsic}}| = g_m r_O = \frac{V_A}{V_T}$$

■ KEY RESULT:

The intrinsic gain $g_m r_O = V_A/V_T$ represents the **maximum** voltage gain achievable by a single transistor. It is approximately **200** for $V_A = 5$ V, and is **independent of bias current!** This is a fundamental property of the device technology.

LECTURE INSIGHT (Lecture 22)

Razavi notes: “The intrinsic gain $g_m r_O$ is the ultimate figure of merit for a transistor as an amplifier. Every circuit we design can only *approach* this limit. In modern integrated circuits, V_A is around 5 V, giving $g_m r_O \approx 200$. If you need more gain, you must cascade stages — a single transistor can never exceed $g_m r_O$.”

3 Input and Output Impedance of the Basic CE Stage

3.1 Input Impedance

lowering both the voltage headroom and the input impedance.

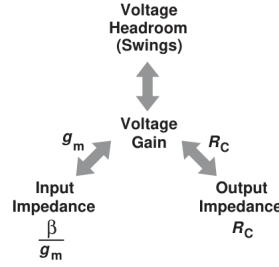


Figure 5.33 CE stage trade-offs.

Figure 5: Figure 5.33 from Razavi — Computing input and output impedance of the CE stage. (a) Input impedance looking into the base, (b) output impedance looking into the collector.

Looking into the base with the emitter grounded:

$$R_{in} = r_{\pi} = \frac{\beta}{g_m} = \frac{\beta V_T}{I_C}$$

Physical intuition: The base current is $i_B = v_{\pi}/r_{\pi}$. Since $v_{\pi} = v_{in}$, the impedance seen at the base is simply r_{π} . For $I_C = 1$ mA and $\beta = 100$: $R_{in} = 100 \times 26$ m/1 m = 2.6 k Ω .

3.2 Output Impedance

To find the output impedance, we set $v_{in} = 0$ and apply a test voltage at the output:

- If $v_{in} = 0$, then $v_{\pi} = 0$, so $g_m v_{\pi} = 0$. The controlled source is off.
- The test source sees only R_C (if $r_O = \infty$) or $R_C \parallel r_O$ (if r_O finite).

$$R_{out} = R_C \quad (\text{without } r_O)$$

$$R_{out} = R_C \parallel r_O \quad (\text{with } r_O)$$

Example 5.21: CE Stage Numerical Analysis (Razavi)

Problem: The circuit of Fig. 5.29 is biased at $I_C = 1$ mA with $R_C = 1$ k Ω . If $\beta = 100$ and $V_A = 10$ V, determine the small-signal voltage gain and the I/O impedances.

Solution:

1. $g_m = I_C/V_T = 1$ mA/26 mV = $(26 \Omega)^{-1} = \mathbf{38.5}$ mA/V
2. $r_O = V_A/I_C = 10$ V/1 mA = $\mathbf{10}$ k Ω
3. Voltage gain: $A_v = -g_m(R_C \parallel r_O) = -38.5 \times (1\text{k} \parallel 10\text{k}) = -38.5 \times 909 = \mathbf{-35}$
4. If $V_A = \infty$: $A_v = -g_m R_C = -38.5 \times 1\text{k} = -38.5 \approx \mathbf{-38}$ (only 8% higher)
5. $R_{in} = r_{\pi} = \beta/g_m = 100/38.5\text{m} = \mathbf{2.6}$ k Ω
6. $R_{out} = R_C \parallel r_O = 1\text{k} \parallel 10\text{k} = \mathbf{909 \Omega}$

EXAM TIP

For exams: The effect of r_O on the basic CE gain is small (typically $< 10\%$) unless R_C is comparable to r_O . If the problem says “ $V_A = \infty$ ” or “neglect the Early effect,” use $A_v = -g_m R_C$. If not stated, include r_O to be safe: $A_v = -g_m(R_C \parallel r_O)$.

4 CE Stage with Emitter Degeneration

4.1 Physical Intuition: Why Add R_E ?

Adding a resistor R_E in series with the emitter creates **negative feedback** that trades gain for several important benefits:

impedance rises to

$$R_{out} = R_C || r_o. \quad (5.134)$$

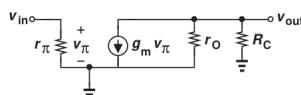


Figure 5.34 CE stage including Early effect

Figure 6: Figure 5.34 from Razavi — (a) CE stage with emitter degeneration. R_E is in the emitter path. (b) Intuitive understanding: R_E “absorbs” part of v_{in} , reducing v_{π} .

The physical story:

1. A positive v_{in} is applied to the base.
2. The emitter voltage also rises (it follows the base voltage, lagging by V_{BE}).
3. The actual voltage across the BE junction is $v_{\pi} = v_{in} - v_E$, which is *smaller* than v_{in} .
4. Therefore $g_m v_{\pi}$ is smaller, and the output voltage swing is reduced.
5. **Result:** The gain is lower, but the amplifier is more linear and stable.

CONCEPT: Negative Feedback in Emitter Degeneration

R_E implements **negative feedback**: if I_C tries to increase, the emitter voltage $V_E = I_E R_E$ rises, which *reduces* $V_{BE} = V_B - V_E$, which brings I_C back down. This self-correcting loop:

- **Stabilizes** the gain against transistor variations
- **Linearizes** the amplifier (less distortion for large signals)
- **Increases** the input impedance
- **Costs:** lower gain magnitude

The “gain-bandwidth tradeoff” and “gain-linearity tradeoff” are central themes of analog design.

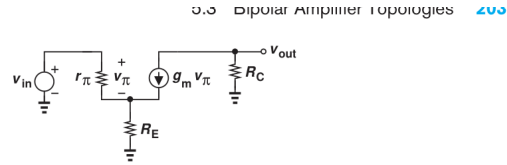


Figure 5.36 Small-signal model of CE stage with emitter degeneration

Figure 7: Figure 5.36 from Razavi — Small-signal model of the CE stage with emitter degeneration. Note that v_π appears across r_π , *not* from the base to ground.

4.2 Small-Signal Analysis of the Degenerated CE Stage

DERIVATION: Voltage Gain with Emitter Degeneration

From the small-signal model (Early effect neglected):

Step 1: KCL at the output node gives:

$$g_m v_\pi = -\frac{v_{out}}{R_C} \quad \Rightarrow \quad v_\pi = -\frac{v_{out}}{g_m R_C}$$

Step 2: Two currents flow through R_E : one from r_π (equal to v_π/r_π) and one from the controlled source ($g_m v_\pi$). The voltage across R_E :

$$v_{RE} = \left(\frac{v_\pi}{r_\pi} + g_m v_\pi \right) R_E = v_\pi \left(\frac{1}{r_\pi} + g_m \right) R_E$$

Step 3: KVL from input to ground:

$$v_{in} = v_\pi + v_{RE} = v_\pi \left[1 + \left(\frac{1}{r_\pi} + g_m \right) R_E \right]$$

Step 4: Combining and using $g_m \gg 1/r_\pi$ (i.e., $\beta \gg 1$):

$$A_v = \frac{v_{out}}{v_{in}} = \frac{-g_m R_C}{1 + g_m R_E}$$

The gain is reduced by a factor of $(1 + g_m R_E)$.

Two Important Limiting Cases

Degeneration: Weak vs. Strong

Regime	Condition	Gain
No degeneration	$g_m R_E = 0$	$A_v = -g_m R_C$
Weak degeneration	$g_m R_E \sim 1$	$A_v = \frac{-g_m R_C}{1 + g_m R_E}$
Strong degeneration	$g_m R_E \gg 1$	$A_v \approx -R_C / R_E$

The strong degeneration case is remarkable: dividing numerator and denominator by g_m :

$$A_v = -\frac{R_C}{\frac{1}{g_m} + R_E} \approx -\frac{R_C}{R_E} \quad (\text{if } g_m R_E \gg 1)$$

■ KEY RESULT:

When $g_m R_E \gg 1$: The gain becomes $A_v \approx -R_C/R_E$, which depends **only on resistor ratios**, not on the transistor parameters (g_m , β , I_C). This is the hallmark of negative feedback: the amplifier becomes insensitive to the transistor's exact characteristics.

LECTURE INSIGHT (Lecture 23)

Razavi explains: “This is a beautiful result. With enough degeneration, the gain is set entirely by passive components — R_C and R_E . The transistor becomes merely the ‘engine’ that provides the current, but the gain depends only on the resistors, which are much more predictable and stable than g_m . This is the power of negative feedback.”

4.3 Input Impedance with Emitter Degeneration

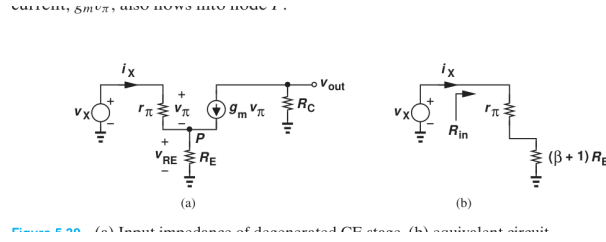


Figure 8: Figure 5.39 from Razavi — (a) Small-signal model for computing R_{in} of the degenerated CE stage. (b) Equivalent circuit showing R_E “multiplied” by $(\beta + 1)$.

DERIVATION: Input Impedance with R_E

Apply a test voltage v_X at the base and measure i_X . Since $v_\pi = r_\pi i_X$, the current through R_E is:

$$i_E = i_X + g_m v_\pi = i_X + g_m r_\pi i_X = i_X (1 + \beta) = (\beta + 1) i_X$$

The voltage drop across R_E is $R_E(\beta + 1)i_X$. By KVL:

$$v_X = r_\pi i_X + R_E(\beta + 1)i_X$$

$$R_{in} = \frac{v_X}{i_X} = r_\pi + (\beta + 1)R_E$$

CONCEPT: Impedance Reflection: The $(\beta + 1)$ Rule

Any impedance Z_E connected between the emitter and ground, when “seen from the base,” appears **multiplied by $(\beta + 1)$** :

$$R_{in,base} = r_\pi + (\beta + 1)Z_E$$

This is because every unit of base current creates $(\beta + 1)$ units of emitter current. The base current “sees” a much larger impedance than actually exists at the emitter.

Why not $r_\pi + R_E$? Because r_π and R_E do *not* carry the same current. The base current i_B flows through r_π , but the emitter current $(1 + \beta)i_B$ flows through R_E . The total voltage is $r_\pi i_B + R_E(1 + \beta)i_B$, yielding the $(1 + \beta)$ scaling on R_E alone.

4.4 Output Impedance with Emitter Degeneration (No r_O)

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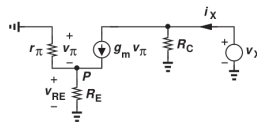


Figure 9: Figure 5.40 from Razavi — Computing output impedance of the degenerated stage by setting $v_{in} = 0$.

Setting $v_{in} = 0$ means $v_{\pi} = 0$ (since v_{π} and v_{RE} must add to zero, and their only driving source is v_{in}). With $v_{\pi} = 0$, the controlled source is off: $g_m v_{\pi} = 0$.

All of the test current i_X therefore flows through R_C :

$$R_{out} = R_C \quad (\text{same as without } R_E, \text{ if } r_O \text{ is neglected})$$

Key insight: Emitter degeneration does **not** change the output impedance when the Early effect is neglected. The output still looks like R_C .

Example 5.24: Designing CE with Degeneration (Razavi)

Problem: A CE stage is biased at $I_C = 1$ mA. The circuit provides a voltage gain of 20 without degeneration and 10 with degeneration. Determine R_C , R_E , and the I/O impedances. Assume $\beta = 100$.

Solution:

1. Without degeneration: $A_v = g_m R_C = 20$
 With $g_m = I_C/V_T = (26 \Omega)^{-1}$: $R_C = 20/g_m = 20 \times 26 = \mathbf{520 \Omega}$
2. With degeneration: $A_v = g_m R_C / (1 + g_m R_E) = 10$
 $20 / (1 + g_m R_E) = 10 \implies 1 + g_m R_E = 2 \implies g_m R_E = 1$
 $R_E = 1/g_m = \mathbf{26 \Omega}$
3. $R_{in} = r_{\pi} + (\beta + 1)R_E = \frac{\beta}{g_m} + 101 \times 26 = 2600 + 2626 \approx \mathbf{5.2 \text{ k}\Omega}$
 (doubled compared to the non-degenerated case!)
4. $R_{out} = R_C = \mathbf{520 \Omega}$

4.5 The Bypass Capacitor Trick

In practice, we often want emitter degeneration for *DC biasing stability* but *not* for small-signal gain. This is achieved by placing a large capacitor C_1 in parallel with R_E :

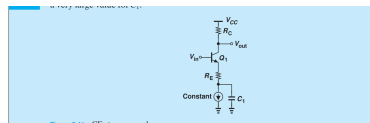


Figure 10: Figure 5.41 from Razavi — CE stage with bypass capacitor. C_1 shorts R_E at signal frequencies, recovering the full gain $-g_m R_C$.

- **At DC:** C_1 is open. R_E is in the circuit and stabilizes the bias (as in Ch. 2).
- **At signal frequencies:** C_1 is a short circuit. R_E is bypassed, and the gain becomes $-g_m R_C$ (full gain, no degeneration).

EXAM TIP

If a problem says “assume C_1 is very large,” immediately replace C_1 with a short circuit for AC analysis. This effectively removes R_E from the small-signal model, giving you the un-degenerated CE gain $A_v = -g_m R_C$.

5 The “Black Box” Perspective: Effective Transconductance G_m

5.1 Motivation

A powerful way to understand degeneration is to treat the *transistor* + R_E combination as a single “super-transistor” with modified parameters:

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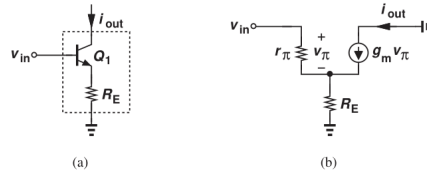


Figure 5.46 (a) Degenerated bipolar transistor viewed as a black box. (b) Small-signal equivalent.

Figure 11: Figure 5.42 from Razavi — (a) The degenerated transistor viewed as a black box (an “active device” with three terminals). (b) Its small-signal equivalent: the controlled source now has a reduced transconductance G_m .

DERIVATION: Effective Transconductance G_m

The output current of the black box in terms of the input voltage:

$$i_{out} = g_m v_\pi = g_m \cdot \frac{v_{in}}{1 + (1/r_\pi + g_m)R_E}$$

For $\beta \gg 1$ (so $g_m \gg 1/r_\pi$):

$$G_m = \frac{i_{out}}{v_{in}} \approx \frac{g_m}{1 + g_m R_E}$$

The voltage gain of the stage with any load R_D is then simply:

$$A_v = -G_m R_D$$

This is elegant: once you know G_m , the gain with *any* load resistance is simply $-G_m R_{load}$. The degenerated transistor behaves like a simpler transistor with a smaller transconductance.

5.2 Effect of Base Resistance

When a resistance R_B appears in series with the base (e.g., the output resistance of the preceding stage, or the Thevenin resistance of the bias network), it **further reduces the gain**:

DERIVATION: Gain with Base Resistance

Using the voltage divider formed by R_B and the input impedance $R_{in} = r_\pi + (\beta + 1)R_E$:

$$v_A = v_{in} \cdot \frac{r_\pi + (\beta + 1)R_E}{r_\pi + (\beta + 1)R_E + R_B}$$

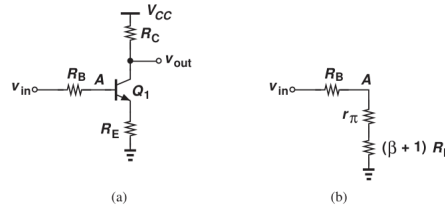


Figure 5.43 (a) CE stage with base resistance (b) equivalent circuit

Figure 12: Figure 5.43 from Razavi — (a) CE stage with base resistance R_B and emitter degeneration. (b) Equivalent circuit showing R_B in series with the input and R_E reflected as $(\beta + 1)R_E$.

The overall voltage gain becomes:

$$A_v = \frac{-\beta R_C}{r_\pi + (\beta + 1)R_E + R_B}$$

Or equivalently:

$$A_v \approx \frac{-R_C}{\frac{1}{g_m} + R_E + \frac{R_B}{\beta + 1}}$$

KEY RESULT:

The base resistance R_B appears in the denominator **divided by** $(\beta + 1)$. This is the “inverse” of the impedance reflection rule: a resistance at the base is “seen from the emitter” as $R_B/(\beta + 1)$.

Design implication: Resistances in series with the emitter and base have similar effects on gain, but the base resistance is scaled down by $\beta + 1$. Both degrade gain, but emitter resistance has a $(\beta + 1)$ times stronger effect.

Chapter 3 Bipolar Amplifiers

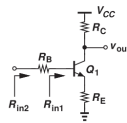


Figure 5.44 Two different input impedances of BJT amplifier

Figure 13: Figure 5.44 from Razavi — Two different input impedances: R_{in1} looking into the base of Q_1 (does not include R_B), and R_{in2} looking into the left terminal of R_B (includes R_B).

6 Output Resistance: The Boosting Effect of Degeneration

6.1 The Key Question

So far we said $R_{out} = R_C$ when neglecting r_O . But when r_O is included, degeneration has a **dramatic** effect on the output impedance seen at the collector of the transistor itself.

6.2 Derivation

mistake here is to write $R_{out} = r_O + R_E$. Since $g_m v_\pi$ flows from the output node into P , resistors r_O and R_E are not in series. We readily note that R_E and r_π appear in parallel, and the current flowing through $R_E \parallel r_\pi$ is equal to i_X . Thus,

$$v_\pi = -i_X(R_E \parallel r_\pi), \quad (5.194)$$

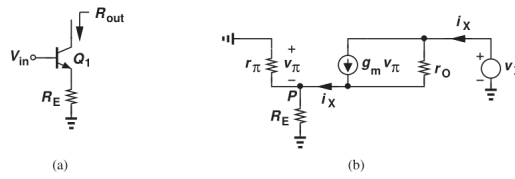


Figure 5.46 (a) Output impedance of degenerated stage. (b) equivalent circuit

Figure 14: Figure 5.46 from Razavi — (a) Degenerated CE stage with output impedance to be determined. (b) Small-signal equivalent: $v_{in} = 0$, R_E and r_π appear in parallel, and the controlled source interacts with r_O .

DERIVATION: Output Impedance with r_O and R_E

Setting $v_{in} = 0$ and applying test voltage v_X at the collector, KVL/KCL analysis gives: Since R_E and r_π are in parallel (from the perspective of the current path):

$$v_\pi = -i_X(R_E \parallel r_\pi)$$

The voltage across r_O is $v_X - v_E$, and the current through r_O has an extra contribution from $g_m v_\pi$. After careful analysis:

$$R_{out,transistor} = [1 + g_m(R_E \parallel r_\pi)] \cdot r_O + R_E \parallel r_\pi$$

Since $g_m r_O \gg 1$:

$$R_{out,transistor} \approx r_O[1 + g_m(R_E \parallel r_\pi)]$$

The circuit output impedance is then:

$$R_{out,circuit} = R_{out,transistor} \parallel R_C$$

The output resistance is **boosted** by a factor of $[1 + g_m(R_E \parallel r_\pi)]$!

Two Important Special Cases

Output Resistance Boosting

Case	Condition	Result
$R_E \gg r_\pi$	$R_E \parallel r_\pi \rightarrow r_\pi$	$R_{out} \approx r_O(1 + g_m r_\pi) \approx \beta r_O$
$R_E \ll r_\pi$	$R_E \parallel r_\pi \rightarrow R_E$	$R_{out} \approx (1 + g_m R_E)r_O$
No degeneration	$R_E = 0$	$R_{out} = r_O$ (no boost)

Maximum possible boost: βr_O . This is the ceiling for a single transistor's output resistance.



Figure 15: Figure 5.47 from Razavi — Stage with r_O drawn explicitly to emphasize that degeneration “boosts” the output resistance seen at the collector.

LECTURE INSIGHT (Lecture 24)

Razavi explains: “The boosting of output resistance by emitter degeneration is one of the most important concepts in analog design. It is the basis for *cascode* circuits, which we’ll study later. A cascode uses one transistor as the degeneration element for another, achieving output resistance of βr_O or even higher. This makes the transistor behave as an almost ideal current source.”

Example 5.28: Designing a Current Source (Razavi)

Problem: Design a current source with $I_C = 1$ mA and $R_{out} = 20$ k Ω . The available transistor has $\beta = 100$ and $V_A = 10$ V. Determine the minimum R_E .

Solution:

1. $r_O = V_A/I_C = 10$ V/1 mA = **10 k Ω**
2. We need $R_{out} = 2r_O$, so the boost factor is 2.
3. Assuming $R_E \ll r_\pi$: $1 + g_m R_E = 2 \implies g_m R_E = 1$
4. $R_E = 1/g_m = V_T/I_C = 26$ mV/1 mA = **26 Ω**
5. Check: $r_\pi = \beta/g_m = 100 \times 26 = 2600$ $\Omega \gg 26$ Ω . $\checkmark$ ($R_E \ll r_\pi$ is valid.)

6.3 Cascode Preview: Transistor as Degeneration

What if instead of a resistor, we use *another transistor* at the emitter? This is the essence of the **cascode** topology:

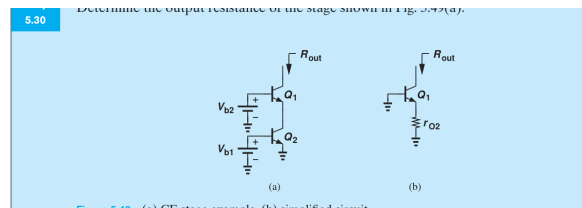


Figure 16: Figure 5.49 from Razavi — (a) A “stack” of two transistors. (b) Q_2 acts as the “degeneration” for Q_1 , and its output resistance r_{O2} replaces R_E in our formula.

Example 5.30: Cascode Output Resistance (Razavi)

Problem: Determine the output resistance of the stage shown in Fig. 5.49(a).

Solution:

Since Q_2 ’s base and emitter are (ac) grounded, Q_2 can be replaced with r_{O2} . This r_{O2} plays the role of the emitter degeneration resistance for Q_1 . By analogy with our formula:

$$R_{out} = [1 + g_{m1}(r_{O2} || r_{\pi1})] \cdot r_{O1}$$

Since $r_{O2} \gg r_{\pi1}$ in typical designs: $R_{out} \approx (1 + g_{m1}r_{\pi1})r_{O1} \approx \beta r_{O1}$.

This is a **massive** improvement: for $\beta = 100$ and $r_{O1} = 10 \text{ k}\Omega$, the cascode gives $R_{out} \approx 1 \text{ M}\Omega$!

7 CE Stage with Biasing: The Complete Amplifier

7.1 The Problem of DC Coupling

In the examples so far, we assumed that v_{in} is directly connected to the base. But in reality, the signal source (e.g., a microphone) can disrupt the DC bias:

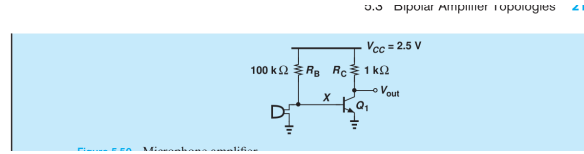


Figure 17: Figure 5.50 from Razavi — A microphone connected directly to the base of a CE amplifier. The microphone's low DC resistance ($< 100 \Omega$) shorts the base bias to near ground.

The problem: A microphone has a very low DC resistance ($< 100 \Omega$). If connected directly to the base, it creates a voltage divider with R_B that pulls V_X to nearly zero, killing the bias point. The transistor turns off!

7.2 The Solution: AC Coupling Capacitors

The fix is to insert a **coupling capacitor** C_1 between the signal source and the base:

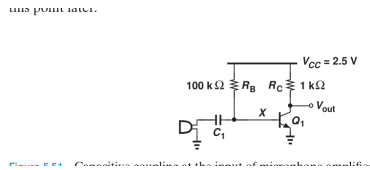


Figure 18: Figure 5.51 from Razavi — Capacitive coupling isolates the DC bias from the signal source. C_1 blocks DC but passes the AC signal.

- **At DC:** C_1 is open. The bias network sets $V_X = R_2 V_{CC} / (R_1 + R_2)$ without interference.
- **At signal frequencies:** C_1 is a short circuit. The signal v_{in} passes through to the base.

7.3 Complete Analysis: Biased CE Stage with Coupling

DERIVATION: Biased CE Stage Analysis

Step 1: DC Analysis — Open all capacitors, kill the signal:

$$I_C = \beta \frac{V_{CC} - V_{BE}}{R_B} \quad (1)$$

$$V_Y = V_{CC} - R_C I_C \quad (2)$$

(For voltage divider bias, use the appropriate formulas from Chapter 2.)

Step 2: Compute small-signal parameters:

$$g_m = I_C / V_T, \quad r_\pi = \beta / g_m, \quad r_O = V_A / I_C$$

Step 3: AC Analysis — Short all capacitors, short V_{CC} to ground:

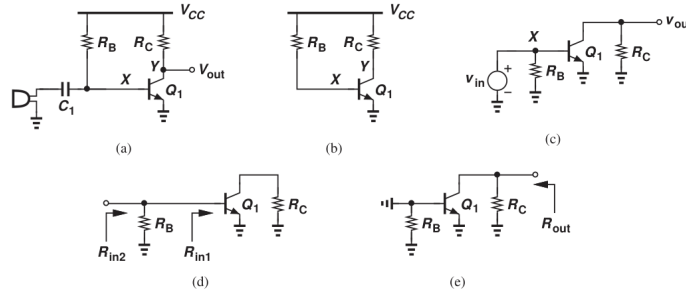
Figure 5.52 (a) Biased CE stage with coupling capacitor. (b) DC equivalent (caps open, signal off). (c) AC equivalent (caps shorted, V_{CC} grounded). (d)/(e) Impedance calculation circuits.

Figure 19: Figure 5.52 from Razavi — (a) Biased CE stage with coupling capacitor. (b) DC equivalent (caps open, signal off). (c) AC equivalent (caps shorted, V_{CC} grounded). (d)/(e) Impedance calculation circuits.

The AC circuit [Fig. 5.52(c)] resembles the basic CE stage, **but R_B appears in parallel with r_π at the input.** This is because R_B connects from the base to V_{CC} , and V_{CC} is AC ground.

Voltage gain from base to output:

$$\frac{v_{out}}{v_X} = -g_m R_C \quad (\text{if } V_A = \infty)$$

Since $v_X = v_{in}$ when v_{in} is an ideal voltage source, the gain is unaffected by R_B . However:

Input impedance:

$$R_{in} = r_\pi \parallel R_B$$

R_B **lowers** the input impedance (appearing in parallel with r_π).

Output impedance:

$$R_{out} = R_C \parallel r_O$$

(unchanged, since setting $v_{in} = 0$ kills the controlled source)

EXAM TIP

Key rule: For AC analysis, any bias resistor connected between a node and V_{CC} (or ground) appears **connected to AC ground**. This means:

- R_B from base to V_{CC} : appears in parallel with r_π in the AC model
- R_1 from V_{CC} to base and R_2 from base to ground: $R_1 \parallel R_2$ appears in parallel with r_π

7.4 Output Coupling and Loading Effects

When a load (e.g., speaker with resistance R_{sp}) is connected at the output through a coupling capacitor C_2 , the AC load becomes $R_C \parallel R_{sp}$:

$$A_v = -g_m (R_C \parallel R_{sp})$$

If $R_{sp} \ll R_C$, the gain drops dramatically. For example, a speaker with $R_{sp} = 8 \Omega$ connected to $R_C = 1 \text{ k}\Omega$ gives:

$$A_v = -g_m (1000 \parallel 8) \approx -g_m \times 7.94$$

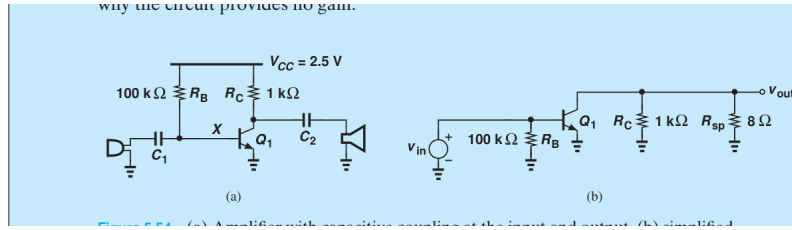


Figure 20: Figure 5.54 from Razavi — (a) Amplifier with coupling capacitors at both input and output. (b) Simplified small-signal model showing the speaker load R_{sp} in parallel with R_C .

This is a gain of less than 1 — useless! The solution is to use a **buffer stage** (Common Collector, Chapter 5) between the amplifier and the load.

Example 5.33: Complete CE Stage Analysis (Razavi)

Problem: The CE stage of Fig. 5.54(a) has $V_{CC} = 2.5\text{ V}$, $R_B = 100\text{ k}\Omega$, $R_C = 1\text{ k}\Omega$, $R_{sp} = 8\text{ }\Omega$. The student measures $V_Y = 1.5\text{ V}$ (collector at quiescent point). If $I_S = 5 \times 10^{-17}\text{ A}$ and $V_A = \infty$:

- Compute β of the transistor.
- Explain why the circuit provides no gain.

Solution:

(a) $V_Y = 1.5\text{ V} \Rightarrow I_C R_C = 2.5 - 1.5 = 1\text{ V} \Rightarrow I_C = 1\text{ mA}$.

$V_{BE} = V_T \ln(I_C/I_S) = 26\text{ mV} \times \ln(10^{-3}/5 \times 10^{-17}) = 26 \times 30.6 = 796\text{ mV} \approx 800\text{ mV}$.

$I_B = (V_{CC} - V_{BE})/R_B = (2.5 - 0.8)/100\text{ k} = 17\text{ }\mu\text{A}$.

$\beta = I_C/I_B = 1\text{ m}/17\text{ }\mu = \mathbf{58.8}$.

(b) The AC total resistance at the collector is $R_C \parallel R_{sp} = 1\text{ k} \parallel 8 = 7.94\text{ }\Omega$.

$|A_v| = g_m(R_C \parallel R_{sp}) = (1\text{ m}/26\text{ m}) \times 7.94 = \mathbf{0.31}$.

The gain is less than unity because the speaker's $8\text{ }\Omega$ resistance dominates the parallel combination.

7.5 Biased CE Stage with Voltage Divider

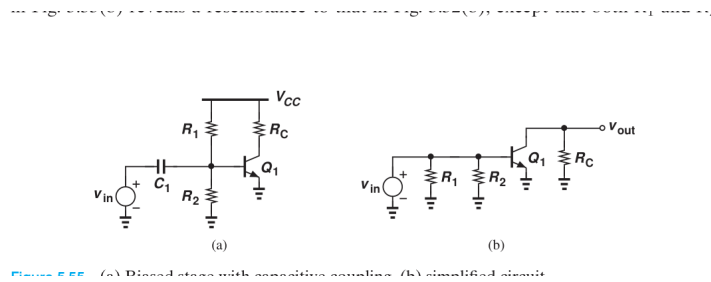


Figure 21: Figure 5.55 from Razavi — (a) Practical CE stage with voltage divider biasing and coupling capacitor. (b) Simplified AC equivalent: $R_1 \parallel R_2$ appears in parallel with r_{π} .

For the most practical biasing topology (voltage divider + R_E), the complete AC analysis gives:

Complete Biased & Degenerated CE Stage**Voltage gain:**

$$A_v = \frac{-g_m R_C}{1 + g_m R_E} \quad (\text{or } -R_C/R_E \text{ if } g_m R_E \gg 1)$$

Input impedance:

$$R_{in} = (R_1 \parallel R_2) \parallel [r_\pi + (\beta + 1)R_E]$$

Output impedance:

$$R_{out} = R_C \parallel r_O$$

If R_E is bypassed by C_E : replace $R_E = 0$ in the gain and R_{in} formulas:

$$A_v = -g_m R_C, \quad R_{in} = (R_1 \parallel R_2) \parallel r_\pi$$

8 CE Amplifier Design Methodology

CONCEPT: Step-by-Step CE Design

Given: V_{CC} , desired gain $|A_v|$, input impedance R_{in} , output impedance R_{out} .

Step 1: Choose I_C .

Higher I_C gives higher g_m (more gain potential) but more power and less r_O .

Step 2: Compute g_m and r_π .

$$g_m = I_C/V_T, \quad r_\pi = \beta/g_m$$

Step 3: Choose R_C .

For the basic CE: $R_C = |A_v|/g_m$ (without degeneration).

Check: $V_{RC} = I_C R_C$ must leave enough headroom ($V_{CE} > V_{BE}$).

Step 4: Choose R_E (if degenerated).

$R_E = R_C/|A_v|$ (for strong degeneration).

Or solve $|A_v| = g_m R_C / (1 + g_m R_E)$ for R_E .

Step 5: Design bias network (R_1, R_2).

- Set $V_X = V_{BE} + I_E R_E$
- Choose $R_1 + R_2$ such that $I_{div} \gg I_B$ (rule: $I_{div} > 10I_B$)
- $R_2/(R_1 + R_2) = V_X/V_{CC}$

Step 6: Verify.

- Active mode: $V_{CE} > V_{BE}$
- Sufficient output swing: $V_{CE,sat} < V_Y < V_{CC}$
- Bias insensitive to β : check with β_{min} and β_{max}

9 Practice Problems

Problem 1: Basic CE Stage

A CE amplifier has $I_C = 0.5$ mA, $R_C = 4$ k Ω , $\beta = 150$, $V_A = 20$ V, $V_{CC} = 5$ V.

- Find g_m , r_π , r_O .
- Find A_v , R_{in} , R_{out} .
- Verify the transistor is in active mode.

Solution:

(a) $g_m = 0.5\text{m}/26\text{m} = \mathbf{19.2}$ mA/V

$r_\pi = 150/19.2\text{m} = \mathbf{7.8}$ k Ω

$r_O = 20/0.5\text{m} = \mathbf{40}$ k Ω

(b) $A_v = -g_m(R_C \parallel r_O) = -19.2\text{m} \times (4\text{k} \parallel 40\text{k}) = -19.2\text{m} \times 3636 = \mathbf{-69.8}$

$R_{in} = r_\pi = \mathbf{7.8}$ k Ω

$R_{out} = R_C \parallel r_O = 4\text{k} \parallel 40\text{k} = \mathbf{3.64}$ k Ω

(c) $V_Y = V_{CC} - I_C R_C = 5 - 0.5\text{m} \times 4\text{k} = 5 - 2 = 3$ V.

$V_{CE} = 3$ V $>$ $V_{BE} \approx 0.75$ V. **Active mode confirmed.**

Problem 2: Emitter Degeneration Design

Design a CE amplifier with $|A_v| = 15$, $I_C = 2$ mA, $\beta = 100$, $V_{CC} = 3.3$ V. Use emitter degeneration. Find R_C , R_E , R_{in} , and verify active mode.

Solution:

$g_m = 2\text{m}/26\text{m} = 76.9$ mA/V.

Method: Use $A_v = -g_m R_C / (1 + g_m R_E)$.

Try strong degeneration: $R_C / R_E = 15 \implies R_C = 15 R_E$.

Need headroom: $V_{CE} > V_{BE} = 0.8$ V and $I_C(R_C + R_E) < V_{CC} - 0.8$.

$2\text{m}(15R_E + R_E) < 2.5 \implies 2\text{m} \times 16R_E < 2.5 \implies R_E < 78$ Ω .

Choose $R_E = 50$ $\Omega \implies R_C = 750$ Ω .

Check: $1 + g_m R_E = 1 + 76.9\text{m} \times 50 = 4.85$.

$A_v = -76.9\text{m} \times 750 / 4.85 = -11.9$. (Not quite 15.)

Adjust: solve $15 = g_m R_C / (1 + g_m R_E)$. With $R_E = 30$ Ω : $1 + g_m R_E = 3.31$.

$R_C = 15 \times 3.31 / 76.9\text{m} = 645$ Ω .

$V_E = I_E R_E \approx 2\text{m} \times 30 = 60$ mV, $V_Y = 3.3 - 2\text{m} \times 645 = 2.01$ V.

$V_{CE} = 2.01 - 0.06 = 1.95$ V $>$ 0.8 V. **Active.** ✓

$R_{in} = r_\pi + 101 \times 30 = 1300 + 3030 = \mathbf{4.33}$ k Ω .

Problem 3: Output Resistance Boosting

A transistor has $g_m = 40$ mA/V, $r_\pi = 2.5$ k Ω , $r_O = 50$ k Ω . It is degenerated with $R_E = 200$ Ω .

- Find R_{out} at the collector (without R_C).
- If $R_C = 5$ k Ω , find the overall circuit R_{out} .
- What is the maximum possible R_{out} at the transistor collector?

Solution:

(a) $R_E \parallel r_\pi = 200 \parallel 2500 = 185$ Ω .

$R_{out} \approx r_O [1 + g_m (R_E \parallel r_\pi)] = 50\text{k} [1 + 40\text{m} \times 185] = 50\text{k} \times 8.4 = \mathbf{420}$ k Ω

(b) $R_{out, \text{circuit}} = 420\text{k} \parallel 5\text{k} = \mathbf{4.94}$ k Ω (dominated by R_C !)

(c) Maximum with $R_E \rightarrow \infty$: $R_{out} \approx r_O (1 + g_m r_\pi) = 50\text{k} (1 + 100) = \mathbf{5.05}$ M Ω

This equals βr_O , confirming the theoretical maximum.

Problem 4: Complete Biased CE Design

Design a complete CE amplifier to achieve $|A_v| = 20$, $R_{in} > 5 \text{ k}\Omega$, using $V_{CC} = 5 \text{ V}$, $\beta = 100$. Use voltage divider bias with R_E . Include coupling capacitors.

Solution:

Choose $I_C = 1 \text{ mA}$. Then $g_m = 38.5 \text{ mA/V}$, $r_\pi = 2.6 \text{ k}\Omega$.

For $|A_v| = 20$ with degeneration: set $g_m R_E = 1$ (moderate degeneration).

$R_E = 1/g_m = 26 \text{ }\Omega$. Then $A_v = g_m R_C / 2 = 20 \implies R_C = 40/g_m = 1.04 \text{ k}\Omega \approx 1 \text{ k}\Omega$.

$R_{in, \text{base}} = r_\pi + 101 \times 26 = 2600 + 2626 = 5226 \text{ }\Omega$.

Design divider: $V_X = V_{BE} + I_E R_E = 0.8 + 1\text{m} \times 26 = 0.826 \text{ V}$.

$V_X/V_{CC} = 0.826/5 = 0.165$. Choose $I_{\text{div}} = 200 \text{ }\mu\text{A}$ ($20 \times I_B$).

$R_1 + R_2 = 5/200\text{ }\mu = 25 \text{ k}\Omega$. $R_2 = 0.165 \times 25\text{k} = 4.1 \text{ k}\Omega$. $R_1 = 20.9 \text{ k}\Omega$.

$R_{in} = (R_1 \parallel R_2) \parallel R_{in, \text{base}} = (20.9\text{k} \parallel 4.1\text{k}) \parallel 5226 = 3.4\text{k} \parallel 5226 = \mathbf{2.06 \text{ k}\Omega}$.

Problem: $R_{in} < 5 \text{ k}\Omega$ due to the bias network! Solution: increase R_1, R_2 (use larger divider resistors) and accept less ideal bias accuracy, or use a bypass capacitor on R_E to boost gain without degeneration: $A_v = -g_m R_C = -38.5$ with $R_C = 520 \text{ }\Omega$, and $R_{in} = (R_1 \parallel R_2) \parallel r_\pi$.

10 Common Mistakes and How to Avoid Them

COMMON MISTAKES

- Mistake:** Forgetting the negative sign in $A_v = -g_m R_C$.

Fix: The CE stage is *always* inverting. If your gain comes out positive, you’ve made a sign error. The physical reason: increasing base voltage increases I_C , which *increases* the drop across R_C , pulling V_{out} *down*.
- Mistake:** Writing $R_{in} = r_\pi + R_E$ (simply adding in series).

Fix: The correct formula is $R_{in} = r_\pi + (\beta + 1)R_E$. The emitter current is $(\beta + 1)i_B$, not i_B , so R_E carries $(\beta + 1)$ times the base current. This “impedance reflection” multiplies R_E by $(\beta + 1)$.
- Mistake:** Claiming $R_{out} = R_C + r_O$ (adding in series).

Fix: r_O and R_C are in **parallel**, not series. Both share the same two nodes (collector and AC ground at V_{CC}). Always: $R_{out} = R_C \parallel r_O$.
- Mistake:** Forgetting that R_B (or $R_1 \parallel R_2$) appears in the AC model.

Fix: In the AC equivalent, V_{CC} is grounded. Any resistor from V_{CC} to the base becomes a resistor from the base to ground, appearing **in parallel** with r_π . This always *lowers* the input impedance.
- Mistake:** Saying degeneration “increases output impedance” without qualification.

Fix: Degeneration increases the output impedance *of the transistor itself* ($r_O \rightarrow [1 + g_m(R_E \parallel r_\pi)]r_O$). But the *circuit* output impedance is this boosted value *in parallel with* R_C : $R_{out,circuit} = R_{out,transistor} \parallel R_C$. If $R_C \ll R_{out,transistor}$, the circuit $R_{out} \approx R_C$ regardless of degeneration.
- Mistake:** Using $A_v = -R_C/R_E$ without checking that $g_m R_E \gg 1$.

Fix: The approximation $A_v \approx -R_C/R_E$ requires strong degeneration ($g_m R_E \gg 1$). For $g_m R_E \sim 1$, use the exact formula $A_v = -g_m R_C / (1 + g_m R_E)$.
- Mistake:** Treating coupling capacitors as short circuits in DC analysis.

Fix: Capacitors are **open** for DC and **short** for AC. Confusing these reverses the entire analysis.

Chapter 3: End-of-Chapter Summary

Core CE Stage

- **Topology:** NPN with R_C to V_{CC} , emitter grounded, input at base.
- **Gain:** $A_v = -g_m R_C$ (inverting). With Early effect: $A_v = -g_m (R_C \parallel r_O)$.
- **Intrinsic gain:** $g_m r_O = V_A/V_T$ — the maximum gain of a single transistor.
- **Input impedance:** $R_{in} = r_\pi = \beta/g_m$.
- **Output impedance:** $R_{out} = R_C$ (or $R_C \parallel r_O$).

Emitter Degeneration

- **Gain:** $A_v = -g_m R_C / (1 + g_m R_E) \approx -R_C / R_E$ for $g_m R_E \gg 1$.
- **Input impedance:** $R_{in} = r_\pi + (\beta + 1)R_E$ (increased by impedance reflection).
- **Output impedance (no r_O):** $R_{out} = R_C$ (unchanged).
- **Output impedance (with r_O):** $R_{out,transistor} = r_O [1 + g_m (R_E \parallel r_\pi)]$ (boosted!).
- **Benefits:** More linear, more stable, gain independent of transistor parameters.
- **Cost:** Lower gain magnitude.

With Base Resistance R_B

- **Gain:** $A_v \approx -R_C / (1/g_m + R_E + R_B/(\beta + 1))$.
- R_B is scaled down by $(\beta + 1)$ in its effect on gain (inverse reflection).

CE with Biasing

- **Coupling capacitors:** Open for DC, short for AC. Isolate bias from signal.
- **Bypass capacitor:** Across R_E — provides degeneration for DC but full gain for AC.
- **Bias resistors in AC model:** $R_1 \parallel R_2$ appears in parallel with r_π , lowering R_{in} .
- **Output coupling + load:** Load in parallel with R_C lowers gain.

Key Design Trade-offs

Increase...	Benefit	Cost
R_C	Higher gain	Less voltage headroom
I_C	Higher g_m (gain)	More power, lower r_O
R_E	Better linearity, stability	Lower gain
$R_1 \parallel R_2$ (bias)	Higher R_{in}	Less accurate bias (more β -sensitive)

“The Common Emitter stage is the building block of all amplifiers. Master it, and every other topology becomes a variation on this theme.” — Razavi